

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2085

University 230 kV
July 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2085. The Interconnection Customer proposes a 81.5 MW battery energy storage facility interconnecting at the University 230 kV substation in the Indiana area. The request was received on May 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 422 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$353,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STA S VAN NUYS – GOULD 230.0 kV	TROY ENERGY LLC – RICHLAND 500.0 kV	105.7	811	6.2
NAPOLEON PEAKING STA – RICHLAND 230.0 kV	TROY ENERGY LLC – RICHLAND 500.0 kV	103.0	1,356	11.9
LAUELES – WARD GULF SUBSTATION 100.0 kV L	DEKALB – JONES CHAPEL 500.0 kV	102.4	371	6.0
WAYNE TRACE – DARLINGTON RD 100.0 kV	RICHLND – DEKALB 500.0 kV line	100.7	702	5.9
BREED – SIDNEY A MURRAY JR HYDROELECTRIC 230.0 kV	DEKALB – RICHLAND SUBSTATION 500.0 kV	100.5	2,568	24.0
LANDSMN CREEK – 24TH & GARDENDALE 230.0 kV L	DEKALB – RICHLAND 500.0 kV Line	100.4	1,003	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STA S VAN NUYS – GOULD 230.0 kV	Reconductor line	\$5,487,000	1% (\$46,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RICHLAND – NAPOLEON PEAKING STATION 230.0 kV	Reconductor line	\$8,844,000	1% (\$63,000)
LAUELES – WARD GULF SUBSTATION 100.0 kV L	Reconductor line	\$1,676,000	1% (\$17,000)
WAYNE TRACE – DARLINGTON ROAD 100.0 kV	Reconductor line	\$668,000	1% (\$4,000)
SIDNEY A MURRAY JR HYDROELECTRIC – BREED 230.0 kV	Reconductor line	\$19,855,000	1% (\$131,000)
LANDSMN CREEK – 24TH & GARDENDALE 230.0 kV L	Reconductor line	\$13,503,000	1% (\$92,000)
	Total allocated to project		\$353,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.